YouTube Comment Analysis

Data source specification & the choice of entity

The dataset is obtained from open source. It contains millions of unlabeled records of YouTube comments for animal-related videos.

Data Source

- GitHub
- Publicly available
- Free and unlimited access



Dataset

- 6 million entries of YouTube comments
- 3 attributes: creator name, user id and comment
- Unlabeled data (i.e. not tagged with classifications)

creator_name	userid	comment
Doug The Pug	87.0	I shared this to ...
Doug The Pug	87.0	Super cute 😀🐕🐶
bulletproof	530.0	stop saying get e...
Meu Zoológico	670.0	Tenho uma jiboia ...
ojatro	1031.0	I wanna see what ...
Tingle Triggers	1212.0	Well shit now Im ...
Hope For Paws - O...	1806.0	when I saw the en...
Hope For Paws - O...	2036.0	Holy crap. That i...
Life Story	2637.0	武器はクエストで貰えるんじゃないん...
Brian Barczyk	2698.0	Call the teddy Larry

Data source specification & the choice of entity

There is a huge volume of YouTube comments available to the public for free. Businesses can apply text analysis to generate revenue from this precious source of data.

Choice of Entity

- YouTube
- One of the largest video-sharing platforms in the world
- Users can upload, view, rate, share, add to playlists, report, comment on videos
- Only registered users can leave comments

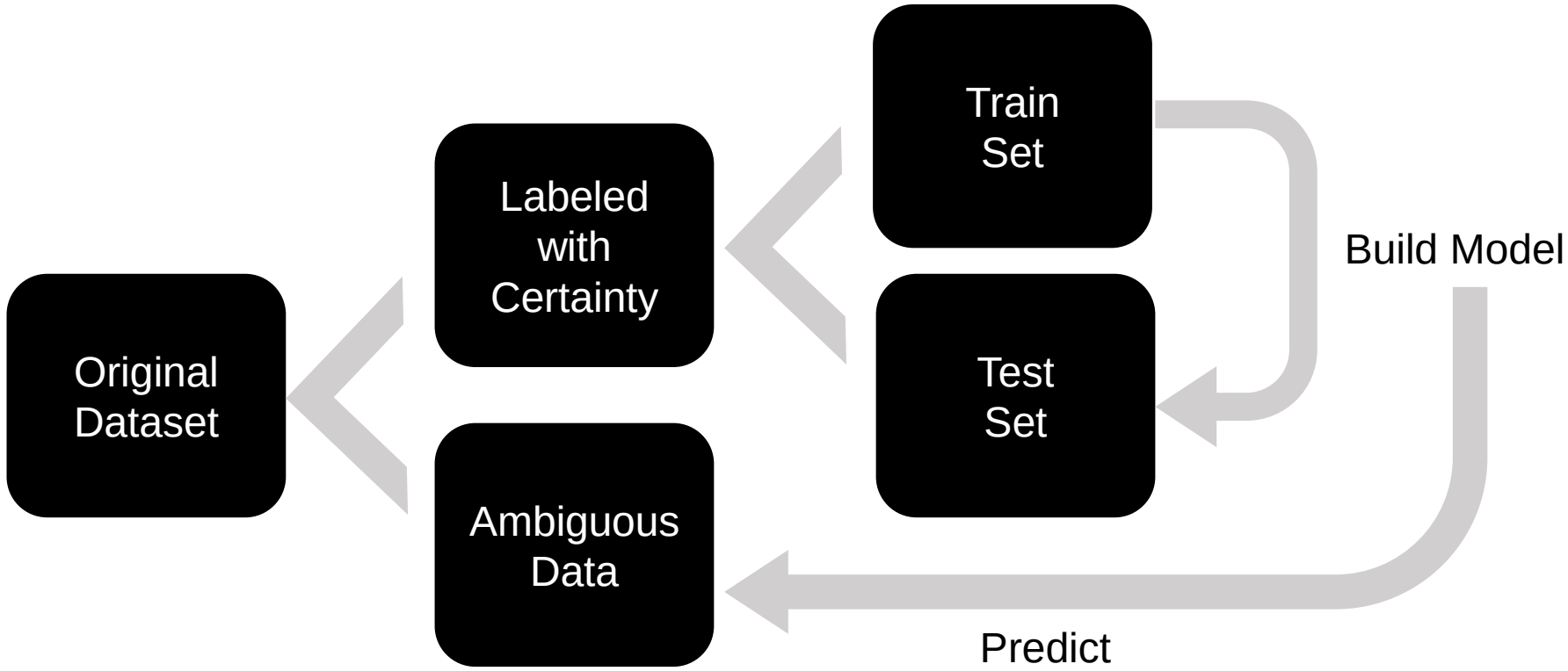
Opportunity

- 500 hours of video content uploaded every minute
- 100 million people make comments to videos every week
- More than 50% of videos include comments from the community

Text Analysis

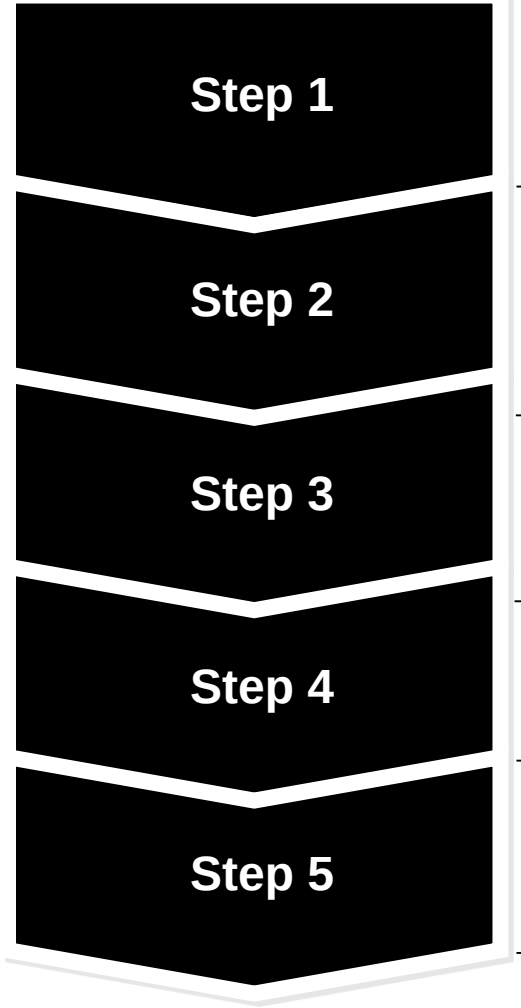
- Identify user segments (cat/dog owner)
- Find out the topics that users are interested in
- Generate insights for marketing and advertising
- Help businesses capitalize on data

Predictive models were built based on the subset of the original data set that was labeled with certainty. The models were then used to predict the rest data.



Design choices and implemented solutions

Since the data is in large volume, I wrote code in databricks notebook to use spark.



	Methods	Implementation
Step 1	• RegexTokenizer • StopWordsRemover • Word2Vec	• Data cleaning • Tokenize the data, remove stopwords and convert to word vectors
Step 2	• Self-defined function • Create subset of data (1m out of 6m) • Down-sampling	• Label dog and cat owners • Get train and test sample
Step 3	• Logistic regression • Random forest	• Train classification models
Step 4	• ROC-AUC • Precision-recall	• Evaluate performance • Use trained model to predict
Step 5	• TF-IDF • LDA	• Find the topics that cat-dog owners care about

Design choices and implemented solutions

Process Data

- Dropped entries with no comments and no userID (about 1% of the original data)
- Defined function to label data
- Labeled cat/dog owners and non-owners
- Get train and test sample (Downsampling to balance the data)

Since there are too many data entries, it is impossible to manually check the information for all users

Build Models

- Tokenized comments
- Removed stop words
- Generated word vectors
- Split into train and test sets
- Built logistic regression and random forest models

Since there were too many negative labels compared to positive labels (100:1), down-sampling was applied to balance the ratio at 2:1 (negative: positive)

The function defined to label cat-dog owners:

```
# find users who own cat or dog
cond = (df_clean["comment"].like("%my dog%") | df_clean["comment"].like("%i have a dog%")\
        | df_clean["comment"].like("%my cat%") | df_clean["comment"].like("%i have a cat%") \
        | df_clean["comment"].like("%my dogs%") | df_clean["comment"].like("%my cats%")\
        | df_clean["comment"].like("%my cat%") | df_clean["comment"].like("%i have dogs%")\
        | df_clean["comment"].like("%i have cats%") | df_clean["comment"].like("%my puppy%")\
        | df_clean["comment"].like("%my kitten%") | df_clean["comment"].like("%i have a puppy%")\
        | df_clean["comment"].like("%i have puppies%"))
```

The results:

creator_name	userid	comment	dog_cat
RaleighLink14	1650352	i feel the same exact way. i own several horses myself and theyre all rescues. my boss even buys all his horses from slaughter houses and from auctions. he either breeds them or puts them in a pasture as retirement or just because of their age. if theyre over 4 or so hell ease them into cutting horse competition. i train and care for all of his horses. for that and being raised as a little girl to love and protect all animals in am all for this movement to protect the wildlife whether domestic or not.	false
Brian Barczyk	2351519	i feel you my dog had to be put down	true
Vet Ranch	2288248	i fell so bad for that dog im crying and i dont cry often	false
Brave Wilderness	2180501	i got a challenge for you try to get bit by a bear and then can you tell us how much it hurt	false
DarkDynastyK9s	968956	i got the same monkey as you 😂😂	false

Get the train/test sample:

```
Number of confirmed user who own dogs or cats:  9229
Number of confirmed user who don't have pet's:  832763
```

- Note that number of negative labels is around 100 times more than positive labels, so here we need to down-sampling the negative labels.
- By rule of thumb, the gap should be no more than 10 times. But here I make them balance to the ratio of 1:2 (positive: negative)

Down-sampling:

```
from pyspark.sql.functions import rand
df_no_pets.orderBy(rand()).createOrReplaceTempView("table")

Num_Pos_Label = df_model.filter(F.col('dog_cat') == True).count()
Num_Neg_Label = df_model.filter(F.col('no_pet') ==  True).count()

#Q1 = spark.sql("SELECT col1 from table where col2>500 limit {}, 1".format(q25))
#pass variable to sql
df_no_pets_down = spark.sql("select * from table where limit {}".format(Num_Pos_Label*2))
```

```
Now after balancing the lables, we have
Positive label:  9229
Negtive label:  18458
```

The train/test sample:

```
+-------+--------------------+-----+
|dog_cat|          wordVector|label|
+-------+--------------------+-----+
|   true|[0.03663111105561...|    1|
|   true|[0.05912114121019...|    1|
|   true|[0.18074695020914...|    1|
|   true|[0.13598554590889...|    1|
|   true|[0.08566602226346...|    1|
|   true|[0.04976289109166...|    1|
|   true|[-0.1439464638630...|    1|
|   true|[-0.0613207606864...|    1|
|   true|[-0.1770278662443...|    1|
|   true|[0.10445102937519...|    1|
+-------+--------------------+-----+
```

Evaluation criteria and results

The Random Forest model generated higher precision and recall, so it was used to classify the rest data.

ROC-AUC

- ROC curve plots true positive rate against false positive rate
- AUC is the area under ROC curve
- Higher AUC indicates better mode

+

Precision-Recall

- Precision measures the accuracy of positive predictions
- Recall measures the coverage of actual positive sample
- Higher precision-recall indicates better model

	Logistic Regression	Random Forest
Accuracy	0.709	0.826
Precision	0.700	0.802
Recall	0.228	0.636
AUC-ROC	0.862	0.908

Findings and conclusions

It is noticeable that text analysis can unearth interesting and unexpected topics the users are interested in. Companies that can capture these interests will be able to benefit from them.

Most important words of topics

```
Topic 1  most important words are : ['cat', 'like', 'get', 'years', 'one', 'love', 'pet', 'turtle', 'got', 'know']
Topic 2  most important words are : ['dog', 'dogs', 'cat', 'boo', 'loves', 'always', 'like', 'love', 'monkey', 'get']
Topic 3  most important words are : ['snakes', 'guinea', 'get', 'ball', 'pig', 'teeth', 'snake', 'im', 'one', 'python']
Topic 4  most important words are : ['dog', 'one', 'like', 'im', 'dogs', 'get', 'love', '3', 'know', 'want']
Topic 5  most important words are : ['cats', 'one', 'ears', 'kitten', 'dog', 'kittens', 'love', 'taylor', 'bath', 'must']
Topic 6  most important words are : ['cat', 'like', 'car', 'get', 'horse', 'back', 'dont', 'dog', 'fucking', 'got']
Topic 7  most important words are : ['dog', 'cat', 'bed', 'like', 'im', 'rip', 'know', 'love', 'meat', 'died']
Topic 8  most important words are : ['horse', 'dog', 'horses', 'im', 'bit', 'time', 'dont', 'also', 'one', 'cat']
Topic 9  most important words are : ['dog', 'dogs', 'people', 'one', 'like', 'killed', 'bit', 'love', 'im', 'gohan']
Topic 10  most important words are : ['bunny', 'dog', 'food', 'like', 'one', 'training', 'cats', 'puppy', 'love', 'videos']
```

Insight:

- Topics that cat-dog owners care about: teeth care, bath, pet food and pet training.
- Companies that offer teeth care, bath, pet food or pet training can target at these users.

Video Creators With Cat And Dog Owners In The Audience (Commented):

creator_name	Number
The Dodo	3141
Taylor Nicole Dean	2563
Brave Wilderness	1988
Brian Barczyk	1887
Hope For Paws - O...	1372
Robin Seplut	1292
Cole & Marmalade	1255
Gohan The Husky	1210
Vet Ranch	1046
Think Like A Horse	1011
Viktor Larkhill	966
Gone to the Snow ...	949
stacyvlogs	948
Zak Georges Dog T...	934
RaleighLink14	669
HammyLux	623
Talking Kitty Cat	532
Happy Tails	492
ViralHog	443
Steff J	432

only showing top 20 rows

Insights:

Companies that provide cat/dog related products can put ads in the videos of these creators.

Future Work:

1. Tune parameters to improve classifier performance.

2. Explore the users' background info and contents of the videos posted to help increase labelling accuracy.

3. Split the cat owners and dog owners and do some exploration.

4. Try to identify more user segments.



Supplement

We then identified cat related topics and dog related topics separately.

Cat related topics:

```
Topic 1  most important words are : ['catnip', 'goes', 'watch', 'white', 'especially', 'first
', 'happy', 'away', 'funny', '11']
Topic 2  most important words are : ['sure', 'box', 'food', 'fine', 'probably', 'birds', 'mak
e', 'cup', 'hes', 'im']
Topic 3  most important words are : ['1', 'sleep', 'beautiful', 'kidney', 'hello', 'healthy',
'give', 'anyone', 'home', '8']
Topic 4  most important words are : ['animals', 'thank', 'home', 'cats', 'year', 'people', 'g
od', 'old', 'lot', 'robin']
Topic 5  most important words are : ['kittens', 'wants', 'hurting', 'allowed', 'ever', 'anyth
ing', 'house', 'act', 'cute', 'makes']
Topic 6  most important words are : ['dog', 'well', 'know', 'sick', 'much', 'fast', 'thats',
'best', 'get', 'cute']
Topic 7  most important words are : ['looks', 'like', 'black', 'sylvester', 'xd', 'welp', 'ga
ve', 'exactly', 'things', 'looked']
Topic 8  most important words are : ['video', 'thinking', 'midnight', 'finally', 'news', 'tv'
, 'extra', 'add', 'smart', 'longer']
Topic 9  most important words are : ['little', 'great', 'one', 'named', 'kitty', 'love', 'lik
e', 'put', 'died', 'cats']
Topic 10  most important words are : ['lol', 'leave', 'us', 'eat', 'thought', 'ate', 'used',
'realise', 'lick', 'chicken']
```

Topics summarized:

- Happy, funny with catnip
- Food, birds
- Sleep, kidney, healthy
- Old people, cat, years
- Kittens, hurting anything in house, allowed
- Dog, much fast, sick
- Love, like, great
- Eat, lick chicken
- Leave us

Supplement

We then identified cat related topics and dog related topics separately.

Dog related topics:

```
Topic 1  most important words are : ['molly', 'put', 'help', 'fight', 'say', 'see', 'died', '
week', 'like', 'looking']
Topic 2  most important words are : ['eat', 'train', 'got', 'feed', 'theyre', 'haha', 'dogs',
'vet', 'easy', 'doge']
Topic 3  most important words are : ['puppy', 'gohan', 'hope', 'better', 'look', 'door', 'get
s', 'form', 'remind', 'fast']
Topic 4  most important words are : ['hes', 'im', 'treat', 'using', 'dogs', 'video', 'great',
'sorry', 'also', 'love']
Topic 5  most important words are : ['one', '2', 'us', 'collar', 'going', 'day', 'dogs', 'awa
y', 'even', 'love']
Topic 6  most important words are : ['home', 'shelby', 'night', 'infections', 'abused', 'rece
ntly', '17', 'previously', '6', 'stopped']
Topic 7  most important words are : ['ok', 'thinks', 'bite', 'happend', 'attack', 'lol', 'sou
nds', 'couldnt', 'notice', 'loyal']
Topic 8  most important words are : ['sad', 'always', 'name', 'people', 'see', 'ive', 'care',
'dogs', 'think', 'looks']
Topic 9  most important words are : ['cat', 'scared', 'sometimes', 'play', 'really', 'used',
'cats', 'new', 'o', 'ago']
Topic 10  most important words are : ['feel', 'teeth', 'different', 'man', 'food', 'wouldnt',
'blind', 'get', 'nice', 'bit']
```

Topics summarized:

- Fight
- Eat, feed, training
- Puppy looks at door with hope (when hosts are away)
- Love, treat greatly
- Abuse, infection
- Loyal, when notice sounds dogs bite
- Dog, people, care, sad
- Cats are scared
- Teeth

Thank You